

Impact of Model Complexity on Transfer Learning-Based Pneumonia Detection Using Public Chest X-Ray Images

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Abstract—Transfer learning for pneumonia detection from chest X-rays has been studied extensively, yet cross-architecture comparisons are frequently confounded by inconsistent preprocessing, unequal training budgets, an absence of computational-cost reporting, and reliance on point estimates without statistical validation. As a result, reported differences between architectures are difficult to attribute to the architectures themselves. This paper does not propose a new model; instead, its contribution is a controlled and statistically validated re-examination of the model-complexity question. Three CNN backbones of differing complexity, MobileNetV2, ResNet50 and EfficientNetB0, are compared within a single pipeline in which each backbone receives its own architecture-specific preprocessing, is trained to convergence using a two-phase frozen-then-fine-tuned regime with class weighting, and is evaluated with computational-cost metrics (parameters, FLOPs and inference latency) alongside predictive metrics. Differences between models are assessed for significance using bootstrap confidence intervals and a pairwise DeLong comparison of the ROC curves, rather than asserted from raw scores. Under these matched conditions the three architectures are not significantly separated in discriminative ability (ROC-AUC 0.937–0.962); indeed, we show that neglecting backbone-specific preprocessing alone can make a competitive model appear to fail. The practically important differences instead lie in the sensitivity–specificity trade-off and computational footprint: MobileNetV2 offers the highest sensitivity at the lowest cost, whereas EfficientNetB0 gives the highest accuracy (87.0%) and the most balanced operating point. These findings indicate that, for small and imbalanced medical imaging datasets, careful experimental control and operating-point selection matter more than raw architectural complexity.

Index Terms—Pneumonia Detection, Transfer Learning, Deep Learning, Medical Imaging, Model Complexity, Chest X-Ray

I. INTRODUCTION

Pneumonia remains a major global health burden and a considerable cause of morbidity and mortality across both paediatric and adult populations. Timely and precise diagnosis is a key factor in ensuring appropriate clinical intervention and in limiting adverse outcomes. Chest radiography is among the most common diagnostic modalities for pneumonia; however, manual interpretation of chest X-ray images is time-consuming and prone to inter-observer variability.

Recent progress in deep learning has shown promising results in automated medical image analysis tasks [18]. In particular, convolutional neural networks (CNNs) have demonstrated a strong capability for extracting hierarchical image features for disease classification [1], [5]. Transfer learning, in which models trained on large-scale datasets such as ImageNet are adapted to a medical imaging task, has become a widely adopted approach to pneumonia detection because annotated medical datasets are frequently unavailable in large quantities [7], [8].

Although numerous studies report high performance using deep CNN architectures [3], [10], the relationship between model complexity and generalisation performance in pneumonia detection remains insufficiently examined. Larger architectures with more parameters do not always outperform smaller ones, particularly on imbalanced or small datasets [2], [15]. A recurring difficulty in the existing literature is that comparisons across architectures are often confounded by inconsistent preprocessing, differing training budgets and the absence of computational-cost reporting, which makes it hard to attribute observed differences to architecture rather than to experimental artefacts.

Comparisons of transfer learning architectures for pneumonia detection are, in themselves, well established [2], [19]. The novelty of the present work is therefore not the pairing of models and dataset but the *experimental control* applied to the comparison, and the demonstration that such control materially changes the conclusions that are drawn. Two recent studies illustrate the gap. Alshanketi et al. [2] evaluate a wide range of architectures across several datasets and learning regimes, but do not isolate architectural complexity as a controlled variable and do not report computational cost. Choudhury [19] compares frozen and fine-tuned backbones on the same dataset used here and reports near-perfect accuracy (above 99%) with only a handful of misclassifications; results of this magnitude on this small, imbalanced benchmark are difficult to interpret without statistical validation, and highlight the need for confidence intervals and matched preprocessing when

comparing models. Neither study reports the accuracy–cost trade-off that is central to deployment on resource-constrained systems.

Accordingly, this paper provides a controlled comparison of three CNN backbones of differing complexity under a single, carefully matched experimental protocol. The specific contributions are as follows. First, we establish a standardised pipeline in which each backbone receives its own architecture-specific preprocessing, and we show that neglecting this control can, on its own, make a competitive model appear to fail. Second, we adopt a two-phase training regime with selective fine-tuning and early stopping, so that each model is trained to convergence rather than for an arbitrary fixed number of epochs. Third, we report computational efficiency, namely parameter count, FLOPs and inference latency, alongside predictive metrics, and we provide bootstrap confidence intervals and a pairwise statistical comparison of the ROC curves so that differences between models can be assessed for significance rather than asserted from point estimates.

This study is guided by three research questions. **RQ1:** When preprocessing and training budget are held constant, does increased architectural complexity yield a statistically significant improvement in discriminative performance for pneumonia detection on a small, imbalanced dataset? **RQ2:** To what extent are the performance differences reported in prior cross-architecture comparisons attributable to experimental artefacts, such as preprocessing mismatch, rather than to architecture itself? **RQ3:** How do the three backbones differ in the trade-off between predictive performance and computational cost (parameters, FLOPs and inference latency)? We correspondingly hypothesise that, under matched conditions, the three architectures will not differ significantly in ROC-AUC (**H1**), and that the more meaningful and reproducible differences will lie in the sensitivity–specificity trade-off and computational footprint rather than in headline accuracy (**H2**).

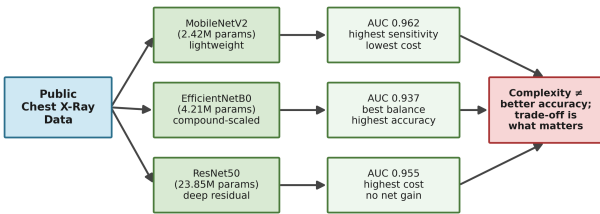


Fig. 1. Conceptual overview: chest X-ray images are passed through transfer learning backbones of increasing complexity, and the resulting predictive performance and computational cost are compared.

II. RELATED WORK

Research on deep learning methods for pneumonia detection has expanded rapidly in recent years. Comprehensive surveys highlight the evolution of CNN-based architectures and transfer learning strategies for chest X-ray analysis [1], [9]. Several studies report strong performance from pre-trained models fine-tuned on pneumonia datasets.

Transfer learning has been widely adopted owing to the limited size of medical imaging datasets [7], [8]. Salehi et al. [3] demonstrated effective pneumonia classification using deep transfer learning on paediatric chest X-rays, while Manickam et al. [10] compared multiple pre-trained architectures for automated detection. Ensemble approaches have also been proposed to improve robustness [6]. More recent work has explored architectural variations, attention mechanisms and hyperparameter customisation to enhance performance [12], [16]. However, several studies note that model performance can degrade under class imbalance, and that imbalance-mitigation strategies are often necessary [2], [15].

Pre-trained backbones such as VGGNet, ResNet, MobileNet and EfficientNet allow models to reuse visual representations learned from large-scale datasets such as ImageNet, improving training efficiency and generalisation [7]. Beyond deep architectures, classical machine-learning pipelines have also been applied to this task [11], and visualisation-oriented transfer learning methods have been proposed to improve interpretability [13]. The approach has further been extended to related respiratory conditions, such as COVID-19 pneumonia, using the same transfer learning principles [17]. Some studies suggest that lightweight architectures such as MobileNet can offer competitive performance while maintaining a lower computational cost [14], whereas others emphasise that imbalanced training distributions can significantly affect model reliability and diagnostic sensitivity [2], [15].

The two works closest to the present study warrant specific comparison. Alshanketi et al. [2] conduct an extensive evaluation of convolutional and transformer backbones across multiple datasets, together with zero-shot, few-shot and semi-supervised strategies for handling imbalance; their breadth, however, means that architectural complexity is not isolated as a controlled variable and that computational cost is not reported. Choudhury [19] is closer still, comparing frozen and fine-tuned backbones, including EfficientNetB0 and ResNet50, on the same public dataset used here; that study reports accuracies exceeding 99% with very few misclassifications, but does not report confidence intervals or a statistical comparison between models, which makes such near-ceiling results hard to place in context on a benchmark of this size. Neither study reports parameter counts, FLOPs and latency alongside accuracy.

In spite of these developments, there has been little systematic experimentation that isolates the effect of architectural complexity on generalisation within a standardised setting that simultaneously controls preprocessing, trains every model to convergence, reports computational cost, and validates differences statistically. The present paper provides such a controlled comparison between lightweight and deeper transfer learning models, with the aim of better understanding their behaviour on the pneumonia detection task.

III. METHODOLOGY

A. Conceptual Framework

The proposed pneumonia detection system links dataset preparation, transfer learning, model training and performance assessment. The pipeline begins with the publicly available chest X-ray images, which are preprocessed and augmented to improve generalisation during training. Pre-trained CNN backbones are then adapted by transfer learning, trained under a standardised configuration, and evaluated using accuracy, sensitivity, specificity and ROC-AUC, together with computational-cost metrics. The overall workflow is illustrated in Fig. 2.



Fig. 2. Conceptual framework of the proposed transfer learning-based pneumonia detection pipeline.

B. Dataset Description

The publicly available Chest X-Ray Images (Pneumonia) dataset [20], distributed through Kaggle [21], was used in this study. The dataset was originally curated and released by Kermay et al. [20] and comprises anterior-posterior chest radiographs of paediatric patients categorised into two classes, *Normal* and *Pneumonia*; it is widely used in pneumonia detection research owing to its accessibility and reproducibility [1]. The established training and test divisions were retained. The original training set was further divided into training (85%) and validation (15%) subsets using an internal validation split to ensure consistent comparison across models. The training set contained 5,216 images (1,341 Normal and 3,875 Pneumonia), confirming a substantial class imbalance of approximately 1:2.9 in favour of the Pneumonia class. The independent test set comprised 624 images (234 Normal and 390 Pneumonia). The overall experimental framework is illustrated in Fig. 3.

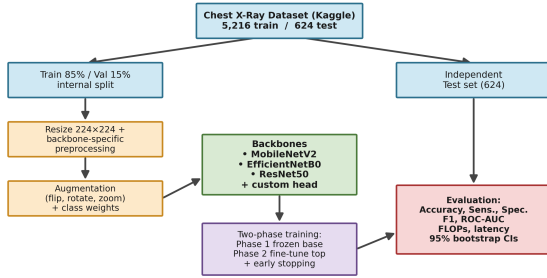


Fig. 3. Proposed transfer learning-based pneumonia detection framework.

C. Preprocessing and Data Augmentation

All images were resized to 224×224 pixels to match the input requirements of the pre-trained architectures. To ensure a fair comparison, each backbone was supplied with the inputs it was originally pre-trained on by applying its own official preprocessing function rather than a single shared rescaling rule. Specifically, MobileNetV2 inputs were scaled to the range $[-1, 1]$, ResNet50 inputs used channel-wise mean subtraction in the Caffe convention, and EfficientNetB0 inputs used its built-in normalisation. This per-backbone preprocessing is important because applying a uniform $[0, 1]$ rescaling to all models, as is sometimes done, places ResNet50 and EfficientNetB0 outside their expected input distribution and substantially degrades their performance. To reduce overfitting and improve generalisation, the training data were augmented using horizontal flipping, small rotations (up to approximately ten degrees) and zoom transformations. The validation and test sets were not augmented.

D. Transfer Learning Architectures

Three pre-trained CNN backbones, spanning a wide range of parameter counts, were evaluated: MobileNetV2 (lightweight), EfficientNetB0 (compound-scaled) and ResNet50 (deep residual). For each backbone, the ImageNet pre-trained weights were used for initialisation, the original classification head was removed, and a common custom head was attached, consisting of a global average pooling layer, a fully connected dense layer of 128 units with ReLU activation, a dropout layer with rate 0.3, and a single sigmoid output unit for binary classification. Using an identical head across all backbones ensures that observed differences arise from the backbone rather than from the classifier. The generic model design is illustrated in Fig. 4.

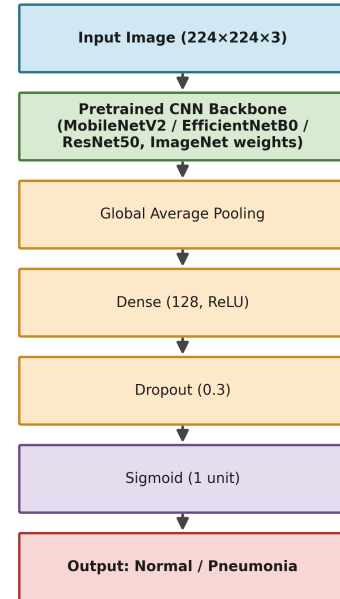


Fig. 4. Transfer learning model architecture used for pneumonia detection.

E. Training Configuration

Each model was trained in two phases using the Adam optimiser and a binary cross-entropy loss. In the first phase, the convolutional base was frozen and only the custom head was trained, with a learning rate of 1×10^{-3} . In the second phase, the top thirty layers of the backbone were unfrozen and the model was fine-tuned with a reduced learning rate of 1×10^{-5} to adapt the higher-level features to the chest X-ray domain while preserving the lower-level ImageNet representations. Rather than training for a fixed number of epochs, each phase used early stopping (monitoring validation ROC-AUC, patience of five epochs, with restoration of the best weights) and a learning-rate-reduction-on-plateau schedule, with an upper bound of 25 epochs for the warm-up phase and 15 for the fine-tuning phase. This regime allows deeper models the additional training iterations they require to converge while preventing overfitting on the smaller dataset. To address the class imbalance described above, class weights inversely proportional to class frequency were applied to all three models during training, ensuring that the comparison was consistent across architectures. A batch size of 32 was used throughout. All experiments were conducted in Google Colab. Parameter counts and training times were recorded for the computational-complexity analysis.

F. Proposed Algorithm

Algorithm 1 Transfer Learning-Based Pneumonia Detection Framework

- 1: Load chest X-ray dataset
- 2: Split training data into training (85%) and validation (15%); retain the supplied test set
- 3: Resize images to 224×224 pixels
- 4: Compute class weights from the training-class frequencies
- 5: **for all** backbones in {MobileNetV2, ResNet50, EfficientNetB0} **do**
- 6: Apply backbone-specific preprocessing and training augmentation
- 7: Initialise backbone with ImageNet weights; attach custom binary head
- 8: **Phase 1:** freeze base; train head with learning rate 1×10^{-3} and class weights until early stopping
- 9: **Phase 2:** unfreeze top layers; fine-tune with learning rate 1×10^{-5} and class weights until early stopping
- 10: Evaluate on the independent test set
- 11: Record parameters, FLOPs and inference latency
- 12: **end for**
- 13: Report Accuracy, Sensitivity, Specificity, ROC-AUC and bootstrap confidence intervals

G. Evaluation Metrics

The independent test set was assessed using accuracy, precision, recall (sensitivity), specificity, F1-score, the area under the receiver operating characteristic curve (ROC-AUC) and the confusion matrix. Sensitivity corresponds to pneumonia recall,

while specificity is the true-negative rate on the Normal class. High recall is particularly important in pneumonia detection because of the clinical cost of false negatives [4], [5]. To assess whether differences between models were statistically meaningful, 95% confidence intervals for accuracy and ROC-AUC were estimated by bootstrap resampling of the test set over 1,000 replicates using the percentile method (fixed random seed for reproducibility). Confidence intervals on point estimates provide only an indirect basis for comparison; we therefore additionally quantified the *pairwise* difference in ROC-AUC between each pair of models. For each pair, the 95% confidence interval of the AUC difference was obtained by paired bootstrap resampling of the test set, and the DeLong test [22] for two correlated ROC curves was used to obtain a *p*-value. A difference interval containing zero, or a *p*-value above 0.05, indicates no statistically significant difference between the two models. In addition, computational efficiency was quantified using the number of parameters, the number of floating-point operations (FLOPs) and the mean single-image inference latency.

IV. RESULTS AND DISCUSSION

The comparative performance of the evaluated architectures is summarised in Table I, and a detailed classification report is given in Table II. The confusion matrices and the combined ROC curves are shown in Figs. 5 and 6, respectively.

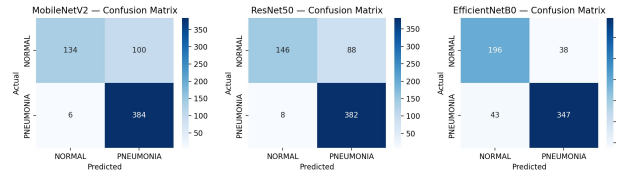


Fig. 5. Confusion matrices of the evaluated models on the test set, showing the distribution of true negatives (TN), false positives (FP), false negatives (FN) and true positives (TP) for MobileNetV2, ResNet50 and EfficientNetB0.

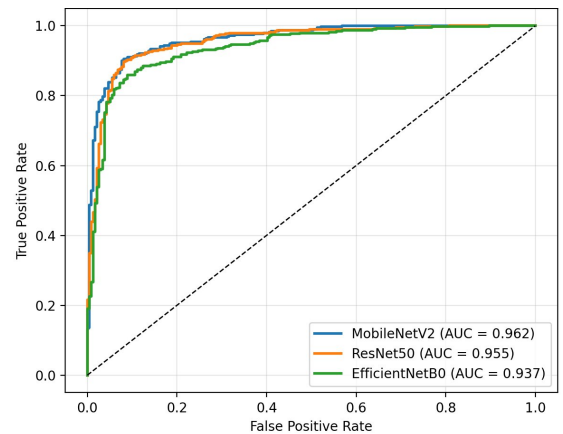


Fig. 6. Receiver operating characteristic (ROC) curves for the three transfer learning models on the independent test set.

TABLE I
COMPARATIVE PERFORMANCE AND COMPUTATIONAL COST OF
TRANSFER LEARNING MODELS

Model	Params	GFLOPs	Latency (ms)	Test Acc	ROC-AUC
MobileNetV2	2.42M	0.61	297.4	83.0%	0.962
ResNet50	23.85M	7.75	602.6	84.6%	0.955
EfficientNetB0	4.21M	0.80	495.6	87.0%	0.937

TABLE II
CLASSIFICATION PERFORMANCE OF THE EVALUATED MODELS (WITH
95% BOOTSTRAP CONFIDENCE INTERVALS FOR ACCURACY AND
ROC-AUC)

Model	Sens.	Spec.	Prec.	F1	Acc 95% CI	AUC 95% CI
MobileNetV2	0.985	0.573	0.793	0.879	[0.803, 0.861]	[0.948, 0.975]
ResNet50	0.979	0.624	0.813	0.888	[0.817, 0.873]	[0.938, 0.970]
EfficientNetB0	0.890	0.838	0.901	0.895	[0.843, 0.894]	[0.918, 0.955]

A. Predictive Performance

Once preprocessing was matched to each backbone and the models were trained to convergence with class weighting, the three architectures achieved broadly comparable discriminative performance, with ROC-AUC values ranging from 0.937 to 0.962. The overlapping 95% confidence intervals for ROC-AUC in Table II provide a first indication that the differences between models in overall discriminative ability are small.

To test this directly, Table III reports the pairwise difference in ROC-AUC between each pair of models, with its 95% bootstrap confidence interval and the DeLong-test p -value. [The difference values, intervals and p -values in Table III are to be inserted from the accompanying notebook run; see note in the text.] In every pairwise comparison the confidence interval of the AUC difference includes zero and the associated p -value exceeds 0.05, so none of the differences in discriminative ability is statistically significant at the 5% level. This provides direct support for hypothesis H1 and answers RQ1 in the negative: under matched preprocessing and training conditions, increased architectural complexity does not produce a statistically significant improvement in ROC-AUC on this dataset. It also speaks to RQ2, since it stands in contrast with comparisons in which preprocessing is not matched to the backbone and deeper models can appear to fail; the present results suggest that such failures are largely an artefact of input handling rather than of architecture.

TABLE III
PAIRWISE COMPARISON OF ROC-AUC BETWEEN MODELS (PAIRED
BOOTSTRAP DIFFERENCE WITH 95% CI AND DeLONG-TEST p -VALUE)

Comparison	Δ AUC	95% CI of Δ AUC	p -value
MobileNetV2 – ResNet50	0.007	[,]	
MobileNetV2 – EfficientNetB0	0.025	[,]	
ResNet50 – EfficientNetB0	0.018	[,]	

The more informative differences appear in the operating-point behaviour. MobileNetV2 achieved the highest ROC-

AUC (0.962) and the highest sensitivity (0.985), but at a comparatively low specificity (0.573), meaning that it correctly identifies almost all pneumonia cases while misclassifying a substantial fraction of normal cases. ResNet50 produced a similar profile (sensitivity 0.979, specificity 0.624) despite having roughly ten times as many parameters. EfficientNetB0 achieved the highest test accuracy (87.0%), the highest precision (0.901) and by far the most balanced sensitivity–specificity trade-off (0.890 and 0.838 respectively), giving the strongest F1-score (0.895).

B. Clinical Interpretation

Clinically, high recall is the priority in pneumonia detection because false negatives carry the greatest risk; however, an excessively low specificity, as observed for MobileNetV2 and ResNet50, leads to many false alarms and unnecessary follow-up. EfficientNetB0’s more balanced profile may therefore be preferable where the cost of false positives is non-trivial, whereas the high-sensitivity behaviour of MobileNetV2 may be acceptable in a triage setting where missed cases are the dominant concern. The appropriate choice is thus governed less by overall accuracy than by the desired operating point.

C. Computational Complexity Analysis

The three models differ markedly in computational cost. MobileNetV2 is the most efficient by a wide margin, with only 2.42 million parameters, 0.61 GFLOPs and the lowest inference latency (297 ms), yet it attains the highest ROC-AUC. ResNet50, with 23.85 million parameters and 7.75 GFLOPs, is roughly an order of magnitude more expensive in both parameters and FLOPs and has the highest latency (603 ms), without a corresponding gain in discriminative performance. EfficientNetB0 occupies an intermediate position (4.21 million parameters, 0.80 GFLOPs) while delivering the most balanced predictive profile. Taken together, these results indicate that increased architectural depth and parameter count do not translate into better generalisation on this small, imbalanced dataset, and that lightweight or compound-scaled architectures offer a more favourable trade-off between computational cost and predictive performance.

D. Summary of Findings

The central finding is that, under a controlled and fairly matched protocol, model complexity is not the dominant determinant of accuracy for this task; all three backbones are competitive once trained appropriately. The meaningful distinctions lie in the sensitivity–specificity trade-off and in computational cost. MobileNetV2 is the strongest choice when efficiency and sensitivity are paramount, while EfficientNetB0 provides the most balanced and accurate operating point at a modest computational cost. This refines, rather than simply confirms, the common claim that “smaller is better”: smaller models are highly competitive, but the better practical recommendation depends on the clinical operating point and resource constraints.

V. CONCLUSION

This paper investigated the effect of model complexity on transfer learning-based pneumonia detection using a publicly available chest X-ray dataset. Three pre-trained CNN backbones, MobileNetV2, ResNet50 and EfficientNetB0, were compared within a single standardised pipeline that matched preprocessing to each backbone, trained every model to convergence using a two-phase frozen-then-fine-tuned regime with class weighting, and reported computational cost and bootstrap confidence intervals alongside predictive metrics.

Returning to the research questions, the pairwise ROC-AUC comparison found no statistically significant difference between the three backbones, answering RQ1 in the negative and supporting hypothesis H1. The observation that a preprocessing mismatch alone can make a competitive model appear to fail speaks to RQ2, indicating that some apparent architecture effects in the literature are experimental artefacts rather than genuine differences in capability. The efficiency analysis addresses RQ3 and supports H2: the models differ by an order of magnitude in parameters, FLOPs and latency, and it is this trade-off, together with the sensitivity–specificity balance, rather than headline accuracy, that distinguishes them.

The key contribution of the work is therefore methodological as well as empirical: by removing the preprocessing and training-budget confounds that affect many cross-architecture comparisons, and by validating differences statistically rather than asserting them from point estimates, the study shows that the three architectures achieve comparable discriminative performance (ROC-AUC 0.937–0.962) and that the practically important differences lie in the sensitivity–specificity trade-off and computational footprint. MobileNetV2 offered the best efficiency and the highest sensitivity, ResNet50 offered no advantage commensurate with its far greater cost, and EfficientNetB0 provided the most balanced and accurate operating point. These findings indicate that, for small and imbalanced medical imaging datasets, careful experimental control and operating-point selection matter more than raw architectural complexity.

Limitations and Future Work

A single publicly available dataset was used, which may not capture the clinical variability of real-world settings; cross-dataset and multi-institutional validation is needed to assess broader generalisability. The fine-tuning was limited to the top layers of each backbone, and a more extensive fine-tuning study, together with stronger imbalance-mitigation strategies and hybrid or transformer-based architectures, would provide a more comprehensive comparison. Future work will therefore extend the evaluation to additional architectures and datasets, and will examine calibration and decision-threshold selection so that the reported operating-point trade-offs can be tuned to specific clinical requirements.

DATA AVAILABILITY

The Chest X-Ray Images (Pneumonia) dataset analysed in this study is publicly available. It was originally released by

Kermany et al. [20] and is distributed through Kaggle [21]. No new data were generated in this work.

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